ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

SPL-I Proposal Report

SWE 4304: Software Project Lab-1

GROUP-15

TITLE OF THE PROJECT

CINE++ — A File-Based Cinema Hall Management System

NAME OF SUPERVISOR

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LIST OF TEAM MEMBERS

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PROJECT OVERVIEW

CINE++ is a console-based Cinema Hall Management System designed to simulate core movie theatre operations through structured programming and efficient file-handling techniques. The system manages movies, showtimes, hall layouts, bookings, pricing

and ticket generation without relying on a database or graphical interface.

The project focuses on creating a modular, maintainable, and fully functional system using C++ and file-based storage. All data including users, movies, halls, seats, tickets, and transaction logs — is stored and processed using text and binary files. The goal is to create a smooth and reliable workflow similar to real cinema ticketing systems while staying within the academic constraints of SPL-1.

MOTIVATION BEHIND THE PROJECT

Cinema ticketing systems are widely used and involve multiple interconnected modules such as scheduling, seat allocation, pricing and user authentication. Recreating this system using only **C++** and file storage presents a strong technical challenge and allows us to explore:

1. Structured and efficient system design
2. Object-oriented programming concepts
3. Realistic scenarios such as dynamic seat updates, automatic ticket creation and smart pricing
4. Designing our own file-based data management system

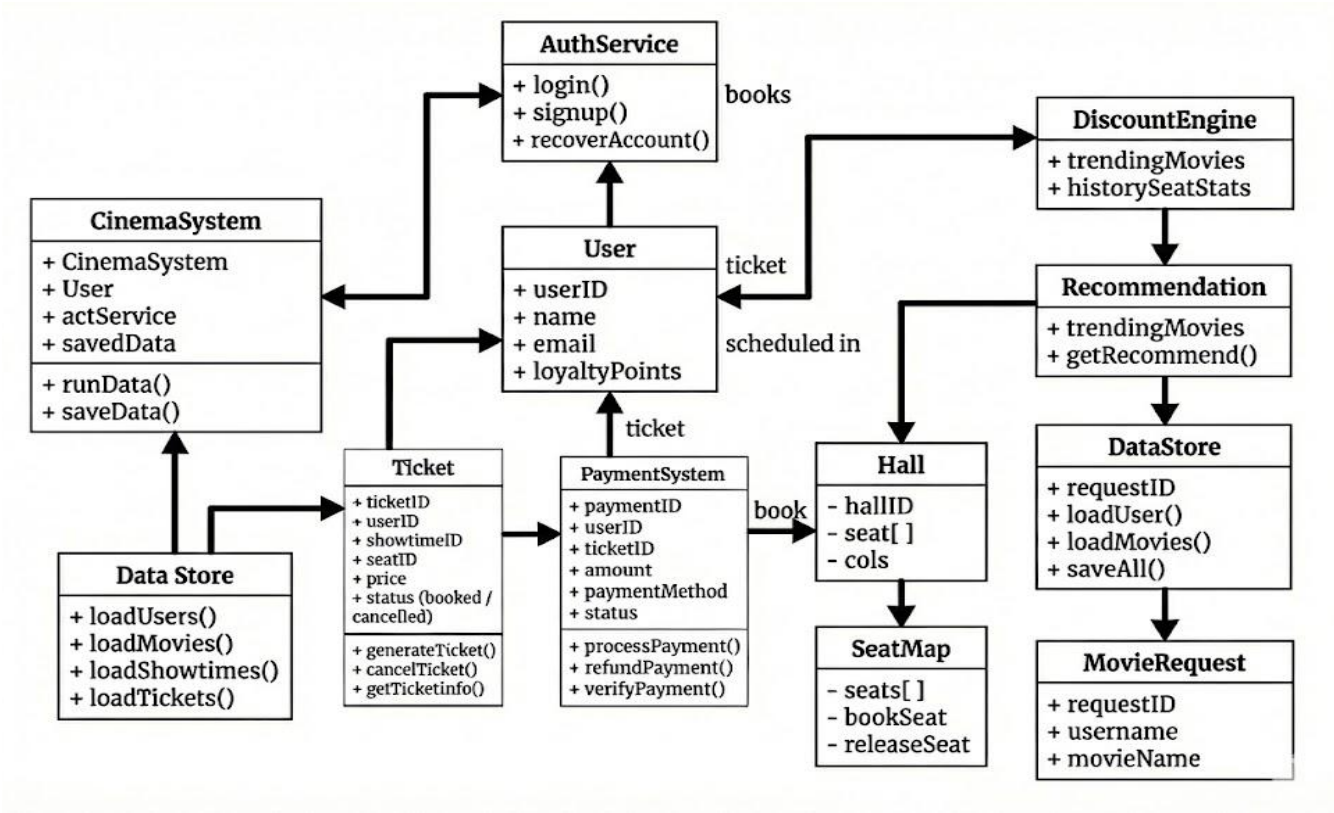
By choosing a practical and relatable real-world domain, this project becomes both educational and engaging, helping us strengthen our programming logic, modular development techniques and problem-solving skills.

FLOWCHART / CLASS DIAGRAM

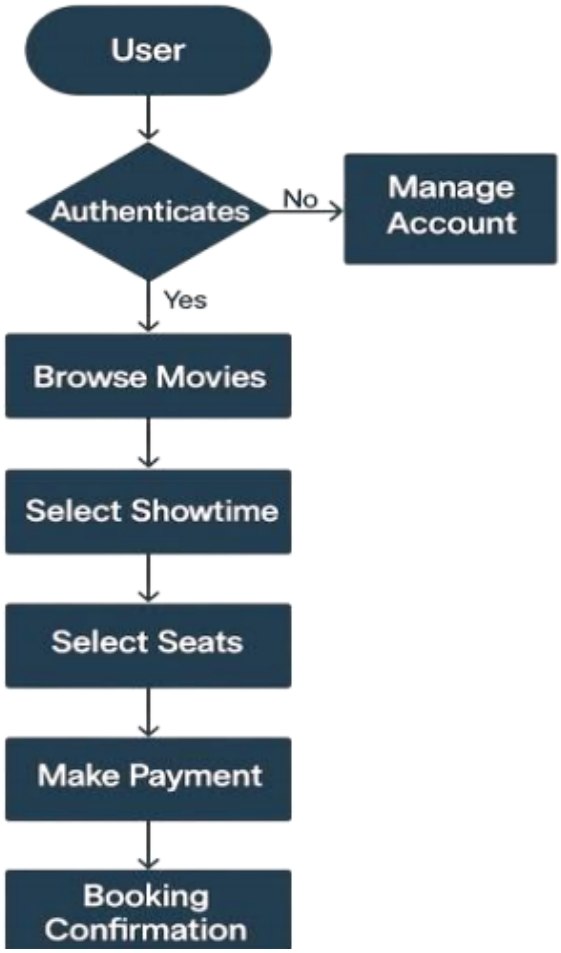
Since some module interactions may evolve during development, the UML will represent the most accurate structure based on current understanding, covering:

- Classes for User, Movie, Show, Hall, Seat, Ticket
- Relationships between modules
- Booking flow from login to seat selection and ticket generation

UML Diagram:



Flowchart:



KEY FEATURES OF THE PROJECT *(USE-CASE DESCRIPTIONS)*

1. User Authentication

- Secure signup and login system
- File-based storage of user credentials
- Basic encryption for passwords
- Role-based access (admin and regular users)

2. Movie & Showtime Management

- Add, edit, view and remove movie information
- Manage genres, runtime, languages and descriptions
- Assign movies to showtimes and halls
- Filter upcoming shows by date and time

3. Hall & Seating System

- Multiple halls with individualized 2D seat layouts
- File-based seat maps for each show
- Real-time updates when seats are booked or released
- Automatic reset after each showtime

4. Booking & Pricing System

- Smart seat selection recommendations
- Group seating and best-seat algorithms
- Dynamic pricing based on time (morning, weekend)
- Discount rules for group bookings and promo codes
- Shopping cart for multi-seat purchases
- Automated billing and price breakdown

5. Ticketing

- Unique ticket ID generation
- Printable ticket format (text-based)
- Validation codes for verification
- Refund/cancellation with automatic seat release

6. Additional Functionalities

- Request new movies
- Movie rating & review storage
- Personalized recommendations based on past bookings
- Admin dashboard for revenue and booking reports
- Basic data security for sensitive information

TOOLS AND TECHNOLOGIES

- **Programming Language:** C++
- **Version Control:** Git and GitHub for collaborative development, version tracking and maintaining clean commit history
- **Storage:** Text and binary files
- **System Architecture:** Modular, object-oriented structure
- **Design Approach:**
    - File-based data management
    - Custom search, sorting, and validation algorithms

- Reusable functions and structured modules

❖ **Interface:** Text-only, console-based interaction

❖ **Development Constraints:**

- No GUI components
- No external libraries for database or graphics
- All functionality developed from scratch

Proposed Timeline

Weeks	Tasks
1–2	Brainstorming, project planning, requirement analysis
3	System outline and architecture design
4	User authentication and file utilities
5	Movie & showtime management module
6	Hall structure and seat grid system
7	Seat allocation logic and best-seat algorithm
8	Dynamic pricing, discounts, and billing
9	Recommendation and review system
10	Ticket generation and refund module
11	Integration of all modules
12	Testing and debugging
13	Documentation preparation
14	Final polishing, adjustments and submission

Action Taken Based on Feedback

- Updated key module descriptions for clarity
- Added Review & Ratings feature as suggested
- Ensured the project aligns with SPL-1 restrictions
- Expanded feature descriptions and system flow for better readability
- Adjusted design to remain fully file-based as required
- Added a clearer breakdown of expected functionalities and modules